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Executive Summary of the Thesis

Title of the thesis

Laurea Magistrale in Xxxxxxxxxxxx Engineering - Ingegneria Xxxxxxxxxxxx

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1. Introduction

Cameras are an essential component of computer vision, and the sensor contained within them are what gives "vision" to computer vision. The precision of numerous different types of sensors has improved substantially over time, but SPAD sensors have distinguished themselves with the ability to bring the precision of imaging down to its base component, the photon. With single-photon avalanche diodes, imaging is now possible in situations such as close to complete darkness, high speed motion, and vast dynamic ranges.

A variety of methods to use the vast amount of data produced by these sensors have been developed in the last 5 years, but most rely on two types of SPAD output, Quanta Image Sensor SPAD and Time Resolved SPAD. Between the two, the most explored has been QIS, based on recording photon detections as a series of binary-valued frames indicating whether at least one photon was detected during each sampling interval. A recent method that makes use of QIS output is Quanta Neural Networks, which proposes a neural network layer capable of interpreting the noisy QIS data and returning frames usable by conventional convolutional layers. However this method, by relying on fixed sampling intervals, does not explicitly exploit the individual photon timestamps available from time-resolved SPAD sensors.

Instead of dividing the acquisition period into sampling intervals and fitting photon detections into fixed timings, TR sensors output the single photon event as a timestamp of when it happened during the exposure time. In this work, i propose a method based on the QNN layers that is capable of using TR-SPAD data, and specifically the information that the timestamps provide.

1.1. Key contributions

The two main contributions of this work are:

- A function for managing a time resolved event stream, based on collecting photon events in bins that preserve timestamp precision.
- A modification of the Sensor-proximal layer proposed by Quanta Neural Networks, adapted to make use of the statistical information in photon interarrival timings.

2. Problem formulation

Consider a simulated TR-SPAD sensor observing a MNIST digit over an acquisition interval

$[0, T_{\text{acq}}]$ [4]. The resulting photon-event stream is represented as

$$\mathcal{S} = (e_i)_{i=1}^{N}, \qquad e_i = (x_i, y_i, t_i),$$

Where N is the number of detected photon-events, $(x_i, y_i) \in \{0, \dots, 27\}^2$ are the pixel coordinates of the event, and t_i is the arrival time measured in seconds

Each recording has a ground-truth digit label $v \in \{0, \dots, 9\}$.

The objective is to develop a CNN-based pipeline that correctly classifies MNIST digits from TR-SPAD event streams and generalizes to previously unseen recordings.

In particular we aim to obtain a correct classification while avoiding separate conventional image-reconstruction methods prior the usage of the CNN. We try to achieve this by directly using temporal information contained in the event stream [2].

3. Related work

3.1. Single-photon imaging and reconstruction

As we discussed above, binary quanta image sensors indicate whether at least one photon was detected during each sampling interval, whereas time-resolved SPADs provide timestamps of individual detections.

An established approach to processing these measurements is to reconstruct scene intensity before applying conventional computer vision algorithms. Ingle et al. [2] demonstrate that inter-photon timing provides information about scene brightness and develop intensity estimators based on the intervals between successive detections. This establishes the value of temporal information for passive single-photon imaging.

Suonsivu et al. [4] introduce TR-MNIST, a simulated dataset of photon detections generated from MNIST images at different illumination levels. Their work provides a benchmark for investigating recognition from photon measurements and motivates approaches that avoid a separate reconstruction stage.

3.2. Streaming integration and changepoint detection

Processing photon measurements sequentially requires a representation that can be updated efficiently while retaining useful temporal information. Integration over longer periods reduces photon noise, but can blur scene changes; shorter integration preserves temporal detail while providing fewer observations.

Zhang et al. [5] address the computational demands of high rate photon streams through a compact representation based on exponential smoothing at multiple predefined temporal scales. The representation is updated incrementally and passed to a neural network for image reconstruction. This demonstrates how temporal aggregation can reduce storage and processing requirements while maintaining access to recent observations.

Bayesian Online Changepoint Detection (BOCPD), introduced by Adams and MacKay [1], provides a probabilistic framework for detecting changes in the statistical properties of a sequence. It maintains a distribution over the run length, representing the time or number of observations since the most recent change. This framework provides the statistical foundation for the adaptive integration used in Quanta Neural Networks.

3.3. Quanta Neural Networks for direct perception

Sundar et al. [3] introduce Quanta Neural Networks (QNNs), which process photon measurements for downstream vision tasks without a separate image reconstruction stage. Their approach incorporates temporal integration layers into existing image and video processing architectures.

The sensor-proximal layer adaptively integrates binary photon measurements using exponential smoothing. Bayesian run-length estimates determine the integration time at each pixel, allowing the accumulation of observations to respond to changes in the photon detection statistics. Intermediate QNN layers use learnable state-space recurrences to aggregate neural-network features over time. Temporal subsampling allows the subsequent network to operate less frequently than the sensor-proximal layer, reducing downstream computation.

4. Sections and subsections

It is convenient to organize the Executive Summary of your thesis into sections and subsec-

tions. If necessary, subsubsections, paragraphs and subparagraphs can be also used. A new section or subsection can be included with the commands

```
\section{Title of the section}

\subsection{Title of the subsection}
```

It is recommended to give a label to each section by using the command

```
\label{sec:section_name}%
```

where the argument is just a text string that you'll use to reference that part as follows: *Section 4 contains* SECTIONS AND SUBSECTIONS ....

5. Equations, Figures, Tables and Algorithms

All Figures, Tables and Algorithms have to be properly referred in the text. Equations have to be numbered only if they are referred in the text.

5.1. Equations

A few important equations related to your work might be reported in the Executive Summary. For example, the Maxwell's equations read:

$$\begin{cases} \nabla \cdot \boldsymbol{D} = \rho, & \text{(1a)} \\ \nabla \times \boldsymbol{E} + \dfrac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{0}, & \text{(1b)} \\ \nabla \cdot \boldsymbol{B} = 0, & \text{(1c)} \\ \nabla \times \boldsymbol{H} - \dfrac{\partial \boldsymbol{D}}{\partial t} = \boldsymbol{J}. & \text{(1d)} \end{cases}$$

Equation (1) is automatically labeled by `cleveref`, as well as Equation (1a) and Equation (1c). Thanks to the `cleveref` package, there is no need to use `\eqref`.

5.2. Figures

To include Figures in your text you can use `TikZ` for high-quality hand-made figures [**?**], or just include them with the command

```
\includegraphics[options]{filename.xxx}
```

where xxx is the format (`.png`, `.jpg`, `.eps`, ...). An example is shown in Figure 1.



Figure 1: Caption of the Figure.

5.3. Tables

Within the environments `table` and `tabular` you can create very fancy tables like the one shown in Table 1.

Example of Table

	column1	column2	column3
row1	1	2	3
row2	α	β	γ
row3	alpha	beta	gamma

Table 1: Caption of the Table.

5.4. Algorithms

Pseudo-algorithms can be written in LaTeX with the `algorithm` and `algorithmic` packages. One example follows.

Algorithm 1 Name of the Algorithm

```
Initial instructions
for for − condition do
    Some instructions
    if if − condition then
        Some other instructions
    end if
end for
while while − condition do
    Some further instructions
end while
Final instructions
```

6. Some further useful recommendations

Theorems and Propositions have to be formatted as follows:

Theorem 6.1. *Write here your theorem.*

Proof. If useful you can report here the proof.

How to write propositions:

Proposition 6.1. *Write here your proposition.*

How to insert itemized lists:

- first item;
- second item.

How to insert numbered lists:

1. first item;
2. second item.

7. Bibliography

The Executive Summary should contain the very essential bibliography of your study. It is suggested to use the BibTeX package [**?**] and save the bibliographic references in the file `bibliography.bib`.

8. Conclusions

A final section containing the main conclusions of your research/study have to be inserted here.

9. Acknowledgements

Here you might want to acknowledge someone.

References

[1] Ryan Prescott Adams and David J. C. MacKay. Bayesian online changepoint detection, 2007.

[2] Atul Ingle, Trevor Seets, Mauro Buttafava, Shantanu Gupta, Alberto Tosi, Mohit Gupta, and Andreas Velten. Passive inter-photon imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8585–8595, 2021.

[3] Varun Sundar, Tianyi Zhang, Sacha Jungerman, and Mohit Gupta. Quanta neural networks: From photons to perception. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 5091–5101, 2025.

[4] Aleksi Suonsivu, Lauri Salmela, Edoardo Peretti, Leevi Uosukainen, Radu Ciprian Bilcu, and Giacomo Boracchi. Time-resolved MNIST dataset for single-photon recognition, 2024.

[5] Tianyi Zhang, Matthew Dutson, Vivek Boominathan, Mohit Gupta, and Ashok Veeraraghavan. Streaming quanta sensors for online, high-performance imaging and vision, 2024.